

Quiz — 1.2.3 Software Development — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (8 marks — 1 each) 1. **B** — Waterfall is linear/sequential, one stage at a time. 2. **B** — Agile delivers iterative increments with continuous client feedback. 3. **B** — XP adds pair programming and constant/test-driven testing to Agile. 4. **B** — Spiral suits large, expensive, high-risk projects via repeated risk analysis. 5. **B** — RAD centres on rapid prototyping refined by user feedback. 6. **B** — Analysis investigates the problem, gathers requirements and checks feasibility. 7. **B** — boundary data lies at the edges of the acceptable range. 8. **B** — Evaluation compares the system against the original requirements.

Section B (12 marks)

9. **[3]** Advantages (1 each, max 2): simple/easy to manage; well documented; clear structure with defined milestones. Disadvantage (1): inflexible / hard to go back and change; the client only sees the finished product at the **end**, so errors in requirements are costly.
10. **[2]** Agile works in iterative increments and gathers continuous client feedback (1), so changing or newly discovered requirements can be built into the next iteration rather than being fixed at the start as in Waterfall (1).
11. **[3]** **Normal/valid** data — typical values within range that should be accepted (1); **boundary** data — values at the edges of the acceptable range (1); **erroneous/invalid** data — values outside the range that should be rejected (1).
12. **[2]** Any two (1 each): design data structures; design algorithms; design the user interface / screen layouts; design inputs, outputs and file/data formats.
13. **[2]** Risk analysis = at each loop the team identifies and evaluates possible risks and plans how to reduce them before continuing (1); on a high-cost project this catches expensive problems early, reducing the chance of costly failure later (1).

Section C (5 marks) 14. **[5]** Recommendation must be justified against the scenario clues. Best answers name **Agile** (or RAD). Up to 5 from: requirements are **unclear/likely to change**, so an iterative methodology that adapts each cycle is suitable (1); **continuous client/user feedback** lets the evolving ideas be incorporated as early versions are shown to real users (1); rapid prototyping/increments deliver working versions quickly for review (1 — especially if RAD named); the **small, skilled team** can cope with the lighter documentation Agile requires (1); fits the **modest budget**/short timescale better than a heavily documented Waterfall, which would lock requirements too early (1). (Reasoned justification with scenario links required for full marks; pure description caps at ~2.)

Total: 8 + 12 + 5 = 25